

Suricata Rules Research

📅 Date July 17, 2026



Rules & Config Research: Commands I Ran:

Checked where the rules which suricata uses for logs actually exist.

```
sudo su
ls -la /var/lib/suricata/rules/
wc -l /var/lib/suricata/rules/suricata.rules
grep "2024364" /var/lib/suricata/rules/suricata.rules
grep "^alert" /var/lib/suricata/rules/suricata.rules | shuf -n 5
sudo grep -A 3 "default-rule-path" /etc/suricata/suricata.yaml
sudo grep -A 5 "rule-files" /etc/suricata/suricata.yaml
suricata-update list-sources
sudo grep -A 5 "eve-log" /etc/suricata/suricata.yaml | head -20
```

```

Terminal - us@ubuntu: ~
File Edit View Terminal Tabs Help
us@ubuntu:~$ sudo ls -la /var/lib/suricata/rules
[sudo: authenticate] Password:
total 43876
drwxr-s--- 2 root    suricata    4096 Jul 15 11:55 .
drwxrws--- 5 suricata suricata    4096 Jul 15 11:55 ..
-rw-r--r-- 1 root    suricata    3228 Jul 15 11:55 classification.config
-rw-r--r-- 1 root    suricata 44913100 Jul 15 11:55 suricata.rules
us@ubuntu:~$ wc -l /var/lib/suricata/rules/suricata.rules
wc: /var/lib/suricata/rules/suricata.rules: Permission denied
us@ubuntu:~$ sudo wc -l /var/lib/suricata/rules/suricata.rules
67948 /var/lib/suricata/rules/suricata.rules
us@ubuntu:~$ sudo grep "2024364" /var/lib/suricata/rules/suricata.rules
alert http $HOME_NET any -> any any (msg:"ET SCAN Possible Nmap User-Agent Observed"; flow:established,to_server; http.user_agent; content:"[20]Nmap"; fast_pattern; classtype:web-application-attack; sid:2024364; rev:5; metadata:affected_product Any, attack_target Client and Server, created at 2017_06_08, deployment Perimeter, performance_impact Low, confidence Medium, signature_severity Informational, updated at 2024_03_07, reviewed at 2024_05_06;)
us@ubuntu:~$ sudo grep "^alert" /var/lib/suricata/rules/suricata.rules | shuf -n 5
alert http $HOME_NET any -> $EXTERNAL_NET any (msg:"ET MALWARE UrgentBot Sending Results to CnC Server (POST)"; flow:established,to_server; urilen:8; http.method; content:"POST"; http.uri; content:"/sendres"; fast_pattern; http.user_agent; content:"WindowsPowerShell"; threshold:type limit,count 1,seconds 300,track by src; reference:md5,be71ca996011e34fe985a0cf674657e; classtype:command-and-control; sid:2063298; rev:1; metadata:affected_product Windows XP_Vista_7_8_10_Server_32_64_Bit, attack_target Client Endpoint, tls_state plaintext, created at 2025_07_03, deployment Perimeter, confidence High, signature_severity Critical, tag c2, updated at 2025_07_03, mitre_tactic_id TA0011, mitre_tactic_name Command And Control, mitre_technique_id T1071, mitre_technique_name Application Layer Protocol; target:src ip;)
alert http $EXTERNAL_NET any -> $HTTP_SERVERS any (msg:"ET WEB SPECIFIC APPS Rialto SQL Injection Attempt -- searchmain.asp cat UNION SELECT"; flow:established,to_server; http.uri; content:"/searchmain.asp?"; nocase; content:"cat="; nocase; content:"UNION"; nocase; pcre:"/UNION\\s+SELECT/i"; reference:cve,CVE-2006-6927; reference:url,www.securityfocus.com/bid/21191; classtype:web-application-attack; sid:2005712; rev:8; metadata:affected_product Web_Server_Applications, attack_target Web_Server, created at 2010_07_30, deployment Datacenter, confidence Medium, signature_severity Major, tag SQL Injection, updated at 2020_09_09, mitre_tactic_id TA0001, mitre_tactic_name Initial Access, mitre_technique_id T1190, mitre_technique_name Exploit Public Facing Application;)
alert dns $HOME_NET any -> any any (msg:"ET MALWARE Win32/Lumma Stealer Related CnC Domain in DNS Lookup (dragoqnfly.run)"; dns.query; bsize:14; content:"dragoqnfly.run"; fast_pattern; nocase; reference:md5,26f0c55c03daf50bf07bebc1d1343707; classtype:domain-c2; sid:2066477; rev:1; metadata:attack_target Client Endpoint, created at 2025_12_26, deployment Perimeter, deployment Internal, malware_family Lumma Stealer, confidence High, signature_severity Major, updated at 2025_12_26, mitre_tactic_id TA0011, mitre_tactic_name Command And Control, mitre_technique_id T1071, mitre_technique_name Application Layer Protocol; target:src ip;)
alert http $HOME_NET any -> $EXTERNAL_NET any (msg:"ET ADWARE PUP Best-targeted-traffic.com Spyware Checkin"; flow:es

```

```

Terminal - us@ubuntu: ~
File Edit View Terminal Tabs Help
rule-files:
- suricata.rules
us@ubuntu:~$ sudo grep -A 5 "rule-files" /etc/suricata/suricata.yaml
rule-files:
- suricata.rules

##
## Auxiliary configuration files.
##
--
#rule-files:
# - firewall.rules

##
## Include other configs
##
us@ubuntu:~$ suricata-update list-sources
17/7/2026 -- 06:58:43 - <Info> -- Using data-directory /var/lib/suricata.
17/7/2026 -- 06:58:43 - <Info> -- Using /usr/share/suricata/rules for Suricata provided rules.
17/7/2026 -- 06:58:43 - <Info> -- Found Suricata version 8.0.6 at /usr/bin/suricata.
17/7/2026 -- 06:58:43 - <Warning> -- Source index does not exist, will use bundled one.
17/7/2026 -- 06:58:43 - <Warning> -- Please run suricata-update update-sources.
Name: abuse.ch/feodotracker
Vendor: Abuse.ch
Summary: Abuse.ch Feodo Tracker Botnet C2 IP ruleset
License: CC0-1.0
Name: abuse.ch/sslbl-blacklist
Vendor: Abuse.ch
Summary: Abuse.ch SSL Blacklist
License: CC0-1.0
Replaces: sslbl/ssl-fp-blacklist
Name: abuse.ch/sslbl-ja3
Vendor: Abuse.ch
Summary: Abuse.ch Suricata JA3 Fingerprint Ruleset
License: CC0-1.0
Replaces: sslbl/ja3-fingerprints
Name: abuse.ch/urlhaus

```

```
Terminal - us@ubuntu: ~
File Edit View Terminal Tabs Help
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, phishing
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-phishing-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, phishing
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: tgreen/hunting
Vendor: tgreen
Summary: Threat hunting rules
License: GPLv3
us@ubuntu:~$ sudo grep -A 5 "eve-log" /etc/suricata/suricata.yaml | head -20
- eve-log:
  enabled: yes
  filetype: regular #regular|syslog|unix_dgram|unix_stream|redis
  filename: eve.json
  # Enable for multi-threaded eve.json output; output files are amended with
  # an identifier, e.g., eve.9.json
--
  # payload-buffer-size: 4 KiB # max size of payload buffer to output in eve-log
  # payload-printable: yes # enable dumping payload in printable (lossy) format
  # payload-length: yes # enable dumping payload length, including the gaps
  # packet: yes # enable dumping of packet (without stream segments)
  # metadata: no # enable inclusion of app layer metadata with alert. Default yes
  # If you want metadata, use:
--
  # payload-buffer-size: 4 KiB # max size of frame payload buffer to output in eve-log
- anomaly:
  # Anomaly log records describe unexpected conditions such
  # as truncated packets, packets with invalid IP/UDP/TCP
  # length values, and other events that render the packet
  # invalid for further processing or describe unexpected
us@ubuntu:~$
us@ubuntu:~$
```

▼ Basic Info about Suricata Rules:

```
sudo grep -A 3 "default-rule-path" /etc/suricata/suricata.yaml
```

This command shows that suricata rules is the file we look for rules. It sets the configuration etc. for suricata.

```
suricata-update list-sources
```

What this command does: asks the `suricata-update` tool to show every available rule *source* it knows about — not just the one you're currently using (Emerging Threats Open), but the full catalog of rule providers that exist in the ecosystem. Some of the sources from the output:

```
Name: abuse.ch/feodotracker
Vendor: Abuse.ch
```

Summary: Abuse.ch Feodo Tracker Botnet C2 IP ruleset
License: CC0-1.0

Name: abuse.ch/sslbl-blacklist
Vendor: Abuse.ch
Summary: Abuse.ch SSL Blacklist
License: CC0-1.0

Name: abuse.ch/sslbl-ja3
...

Name: abuse.ch/urlhaus
Vendor: abuse.ch
Summary: Abuse.ch URLhaus Suricata Rules
License: CC0-1.0

Name: Aleksibovellan/nmap
Vendor: Aleksibovellan
Summary: Suricata IDS/IPS Detection Rules Against NMAP Scans
...

Name: stamus/nrd-phishing-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, phishing
License: Commercial

Name: tgreen/hunting
Vendor: tgreen
Summary: Threat hunting rules
License: GPLv3

Eve-log is the file that does the conf. etc. of eve.json.

What this proves: this is the exact configuration block controlling everything about `eve.json` — confirming `enabled: yes` (it's actually turned on), `filetype: regular` (writing to a normal file, not sending to syslog/Redis/etc., though those are supported alternatives), and `filename: eve.json` (the actual filename). This directly explains **where your alert data has been coming from this whole time** — this config block is the reason `/var/log/suricata/eve.json` exists and gets continuously written to.

▼ Summary

1. `suricata.yaml` is the master config file, telling Suricata two critical things: (a) load rules from `/var/lib/suricata/rules/suricata.rules`, and (b) write all output to `eve.json` (plus `fast.log`)
2. `suricata.rules` is the actual compiled ruleset — currently ~68,000 lines/rules, downloaded from **Emerging Threats Open** via `suricata-update`, each rule following the syntax we broke down (action, protocol, direction, content match, classification, metadata, MITRE ATT&CK mapping)
3. `suricata-update list-sources` reveals this is just one of *many* possible rule providers in the ecosystem — free (Abuse.ch, community GitHub rules) and commercial (Stamus Networks) — showing genuine breadth in how Suricata's detection capability can be extended
4. **The full pipeline:** traffic hits the network interface → gets checked against every rule in `suricata.rules` → matches get logged to both `fast.log` (quick summary) and `eve.json` (full structured detail) → your custom-built live dashboard reads and displays `eve.json` in real time

▼ The SID System

Reserved Ranges of SID: <https://sidallocation.org/>

SID = Signature ID, and it's genuinely the backbone of how Suricata (and Snort, its predecessor) organizes rules globally.

SID numbering ranges (this is a real, documented convention)

- **1–1,000,000:** reserved for Snort's own official rules (historically)
- **2,000,000+:** reserved for Emerging Threats rules
- **Local/custom rules:** convention says use SIDs **1,000,000+** for your own custom-written rules, to avoid colliding with any public ruleset

This is why your Nmap rule was `sid:2024364` — that `2,000,000+` number instantly signals "this is an Emerging Threats rule," even before you check the vendor field.

Does the SID match the original source's ID (e.g., a CVE number)?

No — SID and CVE are two separate numbering systems. SID is Suricata/Snort's *internal* tracking number, assigned by whoever wrote the rule (Emerging Threats, in this case). CVE is a completely separate, industry-wide vulnerability identifier (Common Vulnerabilities and Exposures, maintained by MITRE).

They get **linked together** via the `reference:` option, though — remember this line from one of the rules you found:

```
reference:cve,CVE-2006-6927;
```

This tells you: "SID 2005712 is Emerging Threat's own internal tracking number for this rule, and it happens to detect exploitation attempts related to the publicly documented vulnerability CVE-2006-6927." So: **SID = ET's internal catalog number. CVE = the public vulnerability database's number.** `reference:` is the bridge connecting the two.

▼ All sources:

```
us@ubuntusur:~$ suricata-update list-sources
17/7/2026 -- 06:58:43 - <Info> -- Using data-directory /var/lib/suricata.
17/7/2026 -- 06:58:43 - <Info> -- Using /usr/share/suricata/rules for Suricata provided rules.
17/7/2026 -- 06:58:43 - <Info> -- Found Suricata version 8.0.6 at /usr/bin/suricata.
17/7/2026 -- 06:58:43 - <Warning> -- Source index does not exist, will use bundled one.
17/7/2026 -- 06:58:43 - <Warning> -- Please run suricata-update update-sources.
Name: abuse.ch/feodotracker
  Vendor: Abuse.ch
  Summary: Abuse.ch Feodo Tracker Botnet C2 IP ruleset
  License: CC0-1.0
Name: abuse.ch/sslbl-blacklist
  Vendor: Abuse.ch
  Summary: Abuse.ch SSL Blacklist
  License: CC0-1.0
  Replaces: sslbl/ssl-fp-blacklist
Name: abuse.ch/sslbl-ja3
  Vendor: Abuse.ch
  Summary: Abuse.ch Suricata JA3 Fingerprint Ruleset
  License: CC0-1.0
  Replaces: sslbl/ja3-fingerprints
Name: abuse.ch/urlhaus
  Vendor: abuse.ch
  Summary: Abuse.ch URLhaus Suricata Rules
  License: CC0-1.0
Name: Aleksibovellan/nmap
  Vendor: Aleksibovellan
  Summary: Suricata IDS/IPS Detection Rules Against NMAP Scans
```

License: MIT
Name: et/open
Vendor: Proofpoint
Summary: Emerging Threats Open Ruleset
License: MIT
Name: et/pro
Vendor: Proofpoint
Summary: Emerging Threats Pro Ruleset
License: Commercial
Replaces: et/open
Parameters: secret-code
Subscription: <https://www.proofpoint.com/us/threat-insight/et-pro-ruleset>
Name: etnetera/aggressive
Vendor: Etnetera a.s.
Summary: Etnetera aggressive IP blacklist
License: MIT
Name: ipfire/dbl
Vendor: IPFire
Summary: IPFire DBL
License: CC-BY-SA-4.0
Name: julioliraup/antiphishing
Vendor: julioliraup
Summary: Antiphishing for protection against phishing attacks
License: GPL-3.0
Name: oisf/trafficid
Vendor: OISF
Summary: Suricata Traffic ID ruleset
License: MIT
Name: pawpatrules
Vendor: pawpatrules
Summary: PAW Patrules is a collection of rules for IDPS / NSM Suricata engine
License: CC-BY-SA-4.0
Name: ptrules/open

Vendor: Positive Technologies
Summary: Positive Technologies Open Ruleset
License: Custom

Name: scwx/enhanced
Vendor: Secureworks
Summary: Secureworks suricata-enhanced ruleset
License: Commercial
Parameters: secret-code
Subscription: <https://www.secureworks.com/contact/> (Please reference CTU Countermeasures)

Name: scwx/malware
Vendor: Secureworks
Summary: Secureworks suricata-malware ruleset
License: Commercial
Parameters: secret-code
Subscription: <https://www.secureworks.com/contact/> (Please reference CTU Countermeasures)

Name: scwx/security
Vendor: Secureworks
Summary: Secureworks suricata-security ruleset
License: Commercial
Parameters: secret-code
Subscription: <https://www.secureworks.com/contact/> (Please reference CTU Countermeasures)

Name: stamus/lateral
Vendor: Stamus Networks
Summary: Lateral movement rules
License: GPL-3.0-only

Name: stamus/nrd-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, complete
License: Commercial
Parameters: secret-code
Subscription: <https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed>

Name: stamus/nrd-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, complete
License: Commercial
Parameters: secret-code
Subscription: <https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed>

Name: stamus/nrd-entropy-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, high entropy
License: Commercial
Parameters: secret-code
Subscription: <https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed>

Name: stamus/nrd-entropy-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, high entropy
License: Commercial
Parameters: secret-code
Subscription: <https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed>

Name: stamus/nrd-phishing-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, phishing
License: Commercial
Parameters: secret-code
Subscription: <https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed>

Name: stamus/nrd-phishing-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, phishing

License: Commercial
Parameters: secret-code
Subscription: <https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed>
Name: tgreen/hunting
Vendor: tgreen
Summary: Threat hunting rules
License: GPLv3

```
us@ubuntu:~$ suricata-update list-sources
17/7/2026 -- 06:58:43 - <Info> -- Using data-directory /var/lib/suricata.
17/7/2026 -- 06:58:43 - <Info> -- Using /usr/share/suricata/rules for Suricata provided rules.
17/7/2026 -- 06:58:43 - <Info> -- Found Suricata version 8.0.6 at /usr/bin/suricata.
17/7/2026 -- 06:58:43 - <Warning> -- Source index does not exist, will use bundled one.
17/7/2026 -- 06:58:43 - <Warning> -- Please run suricata-update update-sources.
Name: abuse.ch/feodotracker
  Vendor: Abuse.ch
  Summary: Abuse.ch Feodo Tracker Botnet C2 IP ruleset
  License: CC0-1.0
Name: abuse.ch/sslbl-blacklist
  Vendor: Abuse.ch
  Summary: Abuse.ch SSL Blacklist
  License: CC0-1.0
  Replaces: sslbl/ssl-fp-blacklist
Name: abuse.ch/sslbl-ja3
  Vendor: Abuse.ch
  Summary: Abuse.ch Suricata JA3 Fingerprint Ruleset
  License: CC0-1.0
  Replaces: sslbl/ja3-fingerprints
Name: abuse.ch/urlhaus
  Vendor: abuse.ch
  Summary: Abuse.ch URLhaus Suricata Rules
  License: CC0-1.0
Name: aleksibovellan/nmap
  Vendor: aleksibovellan
  Summary: Suricata IDS/IPS Detection Rules Against NMAP Scans
  License: MIT
Name: et/open
  Vendor: Proofpoint
  Summary: Emerging Threats Open Ruleset
  License: MIT
Name: et/pro
  Vendor: Proofpoint
  Summary: Emerging Threats Pro Ruleset
  License: Commercial
  Replaces: et/open
  Parameters: secret-code
  Subscription: https://www.proofpoint.com/us/threat-insight/et-pro-ruleset
```

```

License: MIT
Name: et/pro
Vendor: Proofpoint
Summary: Emerging Threats Pro Ruleset
License: Commercial
Replaces: et/open
Parameters: secret-code
Subscription: https://www.proofpoint.com/us/threat-insight/et-pro-ruleset
Name: etnetera/aggressive
Vendor: Etnetera a.s.
Summary: Etnetera aggressive IP blacklist
License: MIT
Name: ipfire/dbl
Vendor: IPFire
Summary: IPFire DBL
License: CC-BY-SA-4.0
Name: julioliraup/antiphishing
Vendor: julioliraup
Summary: Antiphishing for protection against phishing attacks
License: GPL-3.0
Name: oisf/trafficid
Vendor: OISF
Summary: Suricata Traffic ID ruleset
License: MIT
Name: pawpatrules
Vendor: pawpatrules
Summary: PAW Patrues is a collection of rules for IDPS / NSM Suricata engine
License: CC-BY-SA-4.0
Name: ptrules/open
Vendor: Positive Technologies
Summary: Positive Technologies Open Ruleset
License: Custom
Name: scwx/enhanced
Vendor: Secureworks
Summary: Secureworks suricata-enhanced ruleset
License: Commercial
Parameters: secret-code
Subscription: https://www.secureworks.com/contact/ (Please reference CTU Countermeasures)

```

```

Subscription: https://www.secureworks.com/contact/ (Please reference CTU Countermeasures)
Name: scwx/malware
Vendor: Secureworks
Summary: Secureworks suricata-malware ruleset
License: Commercial
Parameters: secret-code
Subscription: https://www.secureworks.com/contact/ (Please reference CTU Countermeasures)
Name: scwx/security
Vendor: Secureworks
Summary: Secureworks suricata-security ruleset
License: Commercial
Parameters: secret-code
Subscription: https://www.secureworks.com/contact/ (Please reference CTU Countermeasures)
Name: stamus/lateral
Vendor: Stamus Networks
Summary: Lateral movement rules
License: GPL-3.0-only
Name: stamus/nrd-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, complete
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, complete
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-entropy-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, high entropy
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-entropy-30-open

```

```

Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-entropy-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, high entropy
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-entropy-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, high entropy
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-phishing-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, phishing
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-phishing-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, phishing
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: tgreen/hunting
Vendor: tgreen
Summary: Threat hunting rules
License: GPLv3

```

▼ Suricata Rules structure

Complete details: <https://docs.suricata.io/en/latest/rules/intro.html>

1. Complete Suricata Rule Structure

Every rule has this overall anatomy:

```
ACTION PROTOCOL SRC_IP SRC_PORT DIRECTION DST_IP DST_PORT (OPTIONS)
```

The Header (before the parentheses)

Component	What it means	Example values
Action	What to do on match	<code>alert</code> , <code>drop</code> , <code>reject</code> , <code>pass</code>
Protocol	Which protocol this rule inspects	<code>tcp</code> , <code>udp</code> , <code>http</code> , <code>dns</code> , <code>tls</code> , <code>ftp</code> , <code>ssh</code> , <code>smb</code> , <code>icmp</code> , <code>ip</code>
Source IP	Where traffic originates	<code>\$HOME_NET</code> , <code>\$EXTERNAL_NET</code> , <code>any</code> , or a literal IP/range
Source Port	Origin port	<code>any</code> , <code>80</code> , <code>[80,443]</code>
Direction	Traffic flow arrow	<code>-></code> (one-way) or <code><></code> (either direction)
Dest IP / Dest Port	Same idea, for the destination	—

The Options (inside the parentheses) — the real detection logic

Options are separated by semicolons. Key categories:

Meta-info options (don't affect matching, just describe the rule):

- `msg:"..."` — human-readable description
- `sid:NNNNNN` — unique Signature ID (critical — covered in depth below)
- `rev:N` — revision number
- `classtype:...` — broad category
- `metadata:...` — free-form extra info (dates, severity, MITRE mapping, etc.)

- `reference:...` — links back to CVEs, URLs, or other identifiers (also critical — covered below)

Payload/content matching options:

- `content:"..."` — exact byte/string pattern to search for
- `nocase` — makes the preceding content match case-insensitive
- `pcr:"/regex/"` — regex-based matching, more flexible than plain `content`
- `depth`, `offset`, `distance`, `within` — control *where* in the payload to search
- `fast_pattern` — marks which content match Suricata should use as its quick initial filter

Flow/state options:

- `flow:established,to_server` — only match on established connections, traffic heading toward the server
- `flowbits:...` — lets rules "remember" state across multiple packets (e.g., "if rule A matched earlier in this session, only then check rule B")

Protocol-specific sticky buffers (Suricata 5+, modern syntax):

- `http.user_agent`, `http.uri`, `http.method`, `dns.query`, `tls.sni`, etc. — restrict a `content` match to a specific structured field rather than the whole raw packet

Threshold/detection-limiting options:

- `threshold:type limit,count 1,seconds 300,track by_src` — e.g., "only alert once per 300 seconds from the same source," to avoid alert-flooding on noisy repeated behavior



The complete alert-triggering mechanism (using my Nmap example)

Here's the exact, step-by-step technical process, from packet arrival to alert:

1. PACKET CAPTURE

Kali sends an HTTP request (as part of nmap -sV's version-detection probing)

→ Packet physically arrives on Suricata VM's ens34 interface (promiscuous mode)

2. DECODING

Suricata's engine parses the raw bytes:

Ethernet frame → IP header → TCP header → HTTP protocol parser

→ Suricata now understands "this is an HTTP request, here's the User-Agent field"

3. PROTOCOL-SPECIFIC BUFFER EXTRACTION

Since this is HTTP, Suricata extracts specific "sticky buffers" –

isolated pieces of the packet, e.g. the http_user_agent buffer

specifically contains just the User-Agent header's value

4. FAST PATTERN MATCHING (performance optimization)

Instead of checking all ~68,000 rules individually against every packet,

Suricata uses the fast_pattern-marked content ("Nmap") as a quick filter

→ builds an internal search index so it can rapidly narrow down

which rules are even worth fully evaluating for this packet

5. FULL RULE EVALUATION

For rules that passed the fast-pattern filter, Suricata now checks

ALL conditions in that rule:

- `flow:established,to_server` → is `this` actually an established connection, headed to a server? **YES**
- `http.user_agent; content:"|20|Nmap"; nocase` → does the User-Agent field contain "`Nmap`" (`case-insensitive`)? **YES**, nmap's tool literally embeds `this in` its User-Agent string during version detection

6. MATCH CONFIRMED → ACTION TRIGGERED

Since **ALL** conditions `in sid:2024364` matched, and `this rule's` action is `"alert"`:
→ Suricata generates an alert event internally

7. LOGGING

The alert gets written simultaneously to:

- `fast.log` (simple one-line summary)
- `eve.json` (full structured **JSON**: timestamp, `src/dest IP/port`, the matched signature, classification, and more)

8. YOUR DASHBOARD

Your custom-built live dashboard polls `eve.json` every second,
sees `this new alert` entry, and displays it `in` the table

▼ Suricata Update:

What `suricata-update` actually does (Behind the Scenes)

When you run `sudo suricata-update`, it kicks off a structured 4-step process:

1. **Downloads the index:** It contacts the central Suricata master index to see what rule sources are available (like Emerging Threats, Abuse.ch, etc.).
2. **Fetches the files:** It downloads the latest rule files (usually `.rules` or `.tar.gz` archives) from all of your *enabled* sources.

3. **Merges and filters:** It parses all the downloaded rules, handles any local modifications you've configured, filters out disabled rules, and merges them all into **one single, massive file:** `/var/lib/suricata/rules/suricata.rules`.
4. **Verifies:** It tests the syntax of the new file to make sure there are no typos or broken rules that would crash Suricata.

▼ More about SID and suricata sources:

1. The Perfect Matches (In our local list & on sidallocation.org)

These are rulesets available locally that map to their officially reserved SID ranges on the website:

Source Name (Local)	Registered Organization (on sidallocation.org)	Reserved SID Range
<code>et/open</code>	Emerging Threats	<code>2,000,000 - 2,103,999</code> & <code>2,400,000 - 2,609,999</code>
<code>et/pro</code>	Emerging Threats	<code>2,800,000 - 2,899,999</code>
<code>oisf/trafficid</code>	OISF (Suricata Engine Events)	<code>2,200,000 - 2,299,999</code>
<code>stamus/</code> (all feeds)	Stamus Networks	<code>3,100,000 - 3,199,999</code>
<code>pawpatrules</code>	Pawpatrules	<code>3,300,000 - 3,399,999</code>
<code>aleksibovellan/nmap</code>	Aleksi Bovellan	<code>3,400,000 - 3,499,999</code>
<code>etnetera/aggressive</code>	Etnetera a.s.	<code>5,000,000 - 5,000,213</code>
<code>julioliraup/antiphishing</code>	Julioliraup	<code>6,000,000 - 6,099,999</code>
<code>ptrules/open</code>	Positive Technologies	<code>10,000,000 - 11,999,999</code>
<code>ipfire/dbl</code>	IPFire	<code>12,000,000 - 12,999,999</code>
<code>tgreen/hunting</code>	Travis Green	<code>2,610,000 - 2,619,999</code>
<code>abuse.ch/</code> (all feeds)	Abuse.ch	<code>902,200,000 - 906,200,096</code>

Other Organizations Registered on SID Allocation (But Not in your Local list)

These are organizations that registered their SID ranges with sidsallocation.org to avoid conflicts, but **do not** have their rulesets integrated into OISF's public `suricata-update` source index.

To use these rules, you would have to download them manually and load them as a custom local file:

- **Corelight** (`3,000,000 - 3,099,999`) — Primarily known for Zeek/Corelight network security sensors.
- **DCSO** (`3,200,000 - 3,299,999`) — German Cybersecurity Organization. They maintain their rulesets on GitHub but don't host them as a default Suricata package source.
- **The Hunters Ledger** (`3,500,000 - 3,599,999`) — A community threat intelligence feed.
- **ExtraHop** (`4,000,000 - 4,099,999`) — Enterprise Network Detection and Response (NDR) company.
- **Wybot SAS** (`9,555,000 - 9,954,999`) — Commercial Wi-Fi security intelligence.
- **jpgview** (`27,990,000 - 27,999,999`) — Custom DNS-over-HTTPS (DoH) detection ruleset.

Local Sources with No Explicit Registered SID Range

These are rule vendors shown in your local `list-sources` output that are **not** explicitly named in the main sidsallocation.org directory:

- `scwx/enhanced` , `scwx/malware` , and `scwx/security` (Secureworks)
 - *Why they are missing:* Secureworks is a major commercial enterprise. While they distribute these rules as commercial add-ons (requiring subscription parameters), they likely deploy rules using custom SIDs or inherit from overlapping commercial blocks not indexed on the open-source working group registry.



Saving files/information about their rules for all sources

▼ Emerging Threat:

Link: <https://rules.emergingthreats.net/>

▼ Link Accuracy:

Is <https://rules.emergingthreats.net/> accurate?

Yes, confirmed, this is genuinely the correct, official Emerging Threats rules distribution site. In fact, look back at your very first `suricata-update` run — it explicitly showed:

```
Fetching https://rules.emergingthreats.net/open/suricata-8.0.6/emerging.rules.tar.gz.
```

That's the exact same domain `suricata-update` already downloaded from there automatically. So you can absolutely go get the raw file directly yourself too, without needing to scrape their web portal.

▼ Downloaded from here:

<https://rules.emergingthreats.net/open/suricata-8.0.6/emerging.rules.tar.gz>

```
Checking https://rules.emergingthreats.net/open/suricata-8.0.6/emerging.rules.tar.gz file
Fetching https://rules.emergingthreats.net/open/suricata-8.0.6/emerging.rules.tar.gz.
```

Complete File Format Reference

1. `.rules` files (majority of files — content-matching type)

Format: Plain text, one Suricata rule per line

```
alert http $HOME_NET any -> any any (msg:"..."; sid:NNNNNN; rev:N; ...)
```

Confirmed via `tor.rules`, and everything we've seen from `suricata.rules` on the VM.

2. `.rules` files (IP-list type) — confirmed via `dshield.rules`

Format: Same rule syntax, but matching on IP addresses/ranges rather than content:

```
alert ip [45.33.41.0/24,66.132.186.0/24,...] any -> $HOME_NET any (msg:"...")
```

Notice the header comment explicitly states its purpose: **Rules to block Dshield identified Top Attackers** (www.dshield.org) — DShield is a well-known, real-time internet attack-tracking project run by the SANS Institute. So `dshield.rules` isn't hand-written detection logic like the Nmap rule — it's a **dynamically generated blocklist** of IPs currently known to be attacking, wrapped in valid Suricata rule syntax.

3. `threatview_CS_c2.rules` — confirmed, same pattern as dshield

Format: Also IP-list style, but sourced from a different threat-intel provider (Threatview.io, tracking Cobalt Strike malware C2 servers specifically). Header confirms: **generated from research from the Threatview.io team**, with the actual source data pulled from

<https://threatview.io/Downloads/High-Confidence-CobaltStrike-C2-Feeds.txt>.

Same wrapper syntax, different upstream data feed. Genuinely current — you can see `updated_at 2026_07_16` (yesterday relative to today), confirming this is a live, actively-refreshed feed, not stale data.

Important nuance to note in your research: this file has a **threshold** clause you haven't seen before:

```
threshold: type limit, track by_src, seconds 3600, count 1
```

This means: "only alert **once per hour** (3600 seconds) per source IP" — a rate-limiting mechanism, since these IP-list rules could otherwise fire repeatedly for the same ongoing malicious connection, flooding your alerts unnecessarily.

4. `compromised-ips.txt` — confirmed, genuinely different format

Format: Not Suricata rule syntax at all — just a raw plain-text list of IP addresses, one per line, no wrapping, no `alert`, no `msg`:

```
100.23.75.120101.200.77.117101.201.226.38
```

This is a **raw data file**, not a rules file — likely referenced/included by `compromised.rules` (which probably does contain the actual `alert ip [$list] ...` wrapper), functioning as a separate, more easily updatable data source that the actual rule file points to.

5. `sid-msg.map` — confirmed format

Format: Pipe-delimited (`||`), three fields per line:

```
SID || Message || Reference(s)
```

Example:

```
2000005 || ET EXPLOIT Cisco Telnet Buffer Overflow || url,www.cisco.com/warp/public/707/cisco-sn-20040326-exploits.shtml
```

Some lines have multiple references chained with additional `||`:

```
2000016 || ET DELETED SSL Bomb DoS Attempt || url,www.microsoft.com/... || CVE,CAN-2004-0120
```

This confirms exactly what we discussed earlier — it's genuinely a lightweight SID → description → reference lookup table, perfect for your compiled index, and it directly shows the **SID-to-CVE bridge** in action (see SID 2000033: `cve,2003-0533`).

Interesting finding: several entries say **"ET DELETED"** in the message (e.g., `2000009 ET DELETED Cisco IOS HTTP DoS`) — meaning this specific SID was later retired/deprecated by Emerging Threats, but its number/history is kept in this map file for reference rather than being deleted outright (good practice — SIDs are never reused for a different rule, even after retirement).

6. `suricata-7.0.3-enhanced-open.txt` — confirmed empty

Just an empty placeholder/marker file — no content to analyze, safe to ignore/note as "empty, no data" in your documentation.

7. `classification.config` — format confirmed from earlier exploration

```
config classification: web-application-attack,Web Application Attack,1
```

8. License files (`BSD-License.txt` , `LICENSE` , `gpl-2.0.txt`)

Plain legal text, standard open-source license terms — no rule data, just governs redistribution rights for different portions of the bundled ruleset.

Summary Table:

File/Type	Format	Purpose
Standard <code>.rules</code> (e.g., <code>tor.rules</code>)	Suricata rule syntax, content-based	Pattern-match specific attack behaviors
IP-list <code>.rules</code> (e.g., <code>dshield.rules</code> , <code>threatview_CS_c2.rules</code>)	Suricata rule syntax, IP-based, often with <code>threshold:</code>	Block/alert on known-bad IP feeds
<code>compromised-ips.txt</code>	Raw IP list, no rule wrapper	Data source feeding into a <code>.rules</code> file

File/Type	Format	Purpose
<code>sid-msg.map</code>	Pipe-delimited: SID Message Reference	Quick SID → description → CVE lookup index
<code>classification.config</code>	CSV-like: name,description,priority	Defines all classtype categories & severities
<code>suricata-7.0.3- enhanced-open.txt</code>	Empty	Placeholder, no content
License files	Plain legal text	Redistribution terms